RAM ROCKETRY

Colorado State University



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About Ram Rocketry

Ram Rocketry is a collegiate rocketry organization at Colorado State University dedicated to firsthand learning through experimental rocketry. We welcome everyone, from seasoned enthusiasts to those simply curious about space. Our members develop high-power rocketry certifications and build real aerospace engineering experience across a range of ambitious projects.

Our mission is to ignite students' interest in rocketry and aerospace, fostering a culture of innovation and perseverance. We are a community of ambitious thinkers driven to take on real engineering challenges, and we are just getting started.

Ram Rocketry extends into **Ram Launch Initiative (RLI)**, our NASA University Student Launch Initiative senior capstone team. Over two years of competition, RLI students designed and flew a full high-power rocket to NASA standards, finishing **7th out of 67 teams** in our debut year (2024).

What We Offer

- Technical Experience through HPR Level 1 & 2 certifications, hands-on experimental rocket design and build cycles.
- Engineering Skills, consisting of MATLAB, Simulink, CAD (SolidWorks/Fusion 360), CFD, FEA, Python, and C++
- Professional Development via networking events and site visits with companies including Lockheed Martin, United Launch Alliance, Sierra Space, and Ursa Major



Our Two-Year Mission Roadmap

Ram Rocketry is executing an ambitious two-year program that builds from a nationally competitive solid-propellant rocket to a fully student-designed liquid propulsion system. These two programs run in parallel, with the IREC 2027 COTS effort serving as the foundation and the Liquid Propulsion Project representing our long-term capability goal.

IREC
INTERNATIONAL ROCKET ENGINEERING COMPETITION

PHASE 1 · IREC 2027 — 10,000 ft COTS Competition

IREC 2027 - 10,000 ft COTS Solid Propellant Rocket

Ram Rocketry will compete for the first time at the Spaceport America Cup (IREC) in June 2027 in the 10,000 ft Commercial Off-The-Shelf (COTS) category. This is a rigorous international student rocketry competition drawing teams from across the globe.

Our vehicle is a student-designed high-power rocket targeting a 10,000 ft apogee using a COTS solid motor. The design and build process spans the full 2026–2027 academic year, incorporating:

- Full rocket design: airframe, nosecone, fins, recovery system, and air-braking system
- Dual-deploy parachute recovery with onboard avionics and flight computer.
- Detailed flight simulations using OpenRocket, openMotor and CFD analysis.
- Competition flight data documentation and technical report submission



Our recent Active Fin Stabilization project ("Long Shot", a two-phase canard control demonstrator) has sharpened our team's avionics and control system capabilities, directly feeding into this competition vehicle's design.

PHASE 2 · IREC 2028 — Liquid Propulsion Rocket

Liquid Propulsion Project - Engine Development (2026–2027)

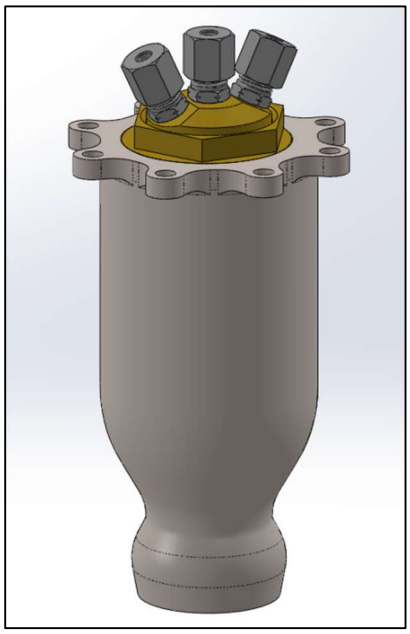
In parallel with the IREC 2027 campaign, our Liquid Propulsion team will spend 2026–2027 designing, manufacturing, and hot-fire testing a student-built bi-propellant liquid rocket engine.

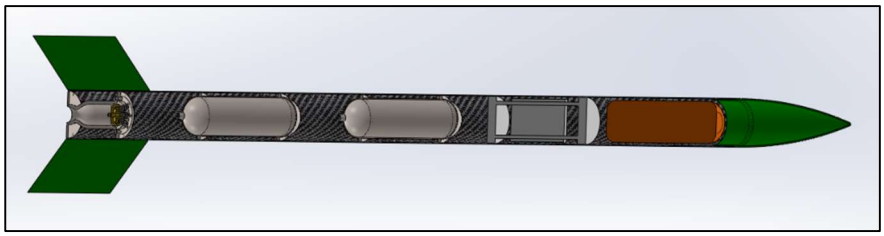
Fuel	Oxidizer
80% Ethanol / 20% H_2O	Nitrous Oxide (N_2O)

- Pintle Injector, proven combustion stability for student-built engines
- Ablative thermal protection; manufacturable in university machine shops
- Pressure-Fed Architecture, reduced system complexity for first development cycle.
- Static Fire Campaign, full test stand development, and hot-fire validation in 2027

IREC 2028 - Liquid-Propelled Rocket Flight

Following engine validation in 2027, the team will integrate the engine into a flight vehicle and compete at IREC 2028, targeting a 10,000 ft apogee. This makes Ram Rocketry one of very few collegiate clubs to fly a student-designed liquid engine at an international competition.





Our Funding Needs

Our two-program roadmap requires sustained investment across two academic years. Below is a summary of how sponsor funds are allocated across each effort.

IREC 2027 - COTS Solid Program

Category	Estimate	Details
Motor and Propulsion	$3,000	COTS solid motor
Airframe & Structure	$6,000	Carbon Fiber, Nosecone, Fins
Avionics & Recovery	$2,000	Flight computers, Parachutes
Travel & Competition Fee	$4,000	Midland, Texas
Miscellaneous & Contingency	$3,000	Testing & Iteration
TOTAL	**$18,000**	

Liquid Propulsion Project - Engine & IREC 2028

Category	Estimate	Details
Test Stand Hardware	$20,000	Plumbing, Valves, Sensors, N_2O/Ethanol
Engine Manufacturing	$12,000	Machining, Ablatives, Fittings
Flight Vehicle	$5,000	Airframe, Recovery, Electronics
Travel & Competition Fee	$3,000	IREC 2028 travel
TOTAL	**$40,000**	

Our combined two-program budget is approximately $51,000. We appreciate and respect sponsors' direct contributions toward a specific program or their interest in providing general support.



Sponsorship Opportunities

Sponsorship is how Ram Rocketry turns ambition into hardware. Every contribution, at any level, directly funds student engineers building and flying real rockets. We are committed to returning genuine value to every sponsor.

Benefit / Tier	SUBORBITAL	LEO	GEO	MEO
Investment	**$1,000+**	**$5,000+**	**$7,500+**	**$10,000+**
Newsletter acknowledgement	✓	✓	✓	✓
Invitation to club events	✓	✓	✓	✓
Feature on website		✓	✓	✓
Feature on social media		✓	✓	✓
Logo on team apparel		✓	✓	✓
Access to resume booklet			✓	✓
Logo on rocket / test stand			✓	✓
Access to design reviews				✓
Invitation to Static Fire				✓
Priority recruiting sessions				✓

For in-kind contributions (materials, machining, software licenses, test equipment) or sponsorship interest enquiries, please contact ramrocketry@gmail.com to discuss how your support can directly serve a specific project need.

To make a monetary donation, visit the QR Code displayed below and enter 'Ram Rocketry' in the Student Organization field, or contact us directly for wire/check arrangements.

Our Sponsors

We are grateful for the organizations that have already invested in Ram Rocketry's growth. Your support fuels our passion and pushes us to take on challenges that matter.

URSA MAJOR
FLY FASTER

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We are honored you are considering partnering with Ram Rocketry.

Together, we will take this rocket higher, and our students even further.

ramrocketry@gmail.com